

2.3 Group 17

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.3 Group 17
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

A student observed the reactions when sodium bromide and sodium fluoride were each reacted separately with concentrated sulfuric acid and concentrated phosphoric acid.

The observations are recorded in the table below.

	Sodium bromide	Sodium fluoride
Conc. H_2SO_4	Steamy fumes and brown vapour	Colourless acidic gas formed
Conc. H_3PO_4	Colourless acidic gas formed	Colourless acidic gas formed

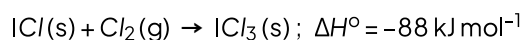
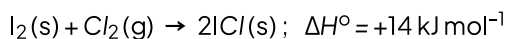
Which deduction can be made from these observations?

- A. Concentrated phosphoric acid is a stronger oxidising agent than concentrated sulfuric acid.
- B. Concentrated phosphoric acid is a stronger oxidising agent than bromine.
- C. Concentrated sulfuric acid is a stronger oxidising agent than fluorine.
- D. Concentrated sulfuric acid is a stronger oxidising agent than bromine.

[1 mark]

Question 2

Iodine trichloride, ICl_3 , is made by reacting iodine with chlorine.



By using the data above, what is the enthalpy change of the formation for solid iodine trichloride?

- A. -162 kJ mol^{-1}
- B. -81 kJ mol^{-1}
- C. -74 kJ mol^{-1}
- D. -60 kJ mol^{-1}

[1 mark]

Question 3

An unlabeled powder is known to be either a single sodium halide or a mixture of two different sodium halides.

To determine what the powder consists of a sample of the powder was first dissolved in water. Then acidified aqueous silver nitrate was added to the new solution, and a pale yellow precipitate was formed. When concentrated aqueous ammonia was added, the precipitate partly dissolved, leaving a darker yellow precipitate.

What might the powder have consisted of?

- A. Sodium iodide only
- B. Sodium bromide only
- C. A mixture of sodium chloride and sodium bromide
- D. A mixture of sodium chloride and sodium iodide

[1 mark]

Question 4

The following two experiments are carried out with anhydrous potassium chloride and observations **X** and **Y** are made at the end of each experiment.

The potassium chloride is dissolved in aqueous silver nitrate, and this is then added to aqueous ammonia- observation **X**.

Concentrated sulfuric acid is added to the potassium chloride, and the fumes produced are bubbled into aqueous potassium iodide solution - observation **Y**.

What are the observations **X** and **Y**?

	X	Y
A	Colourless solution	Brown solution
B	Colourless solution	Colourless solution
C	White precipitate	Brown solution
D	White precipitate	Colourless solution

[1 mark]

Question 5

Element 85, astatine, is a halogen. If concentrated sulfuric acid is added to sodium astatide the resultant products are a mix of astatine, hydrogen astatide, hydrogen sulfide and sodium sulfate.

Which product is formed by the oxidation of one of the constituents of sodium astatide?

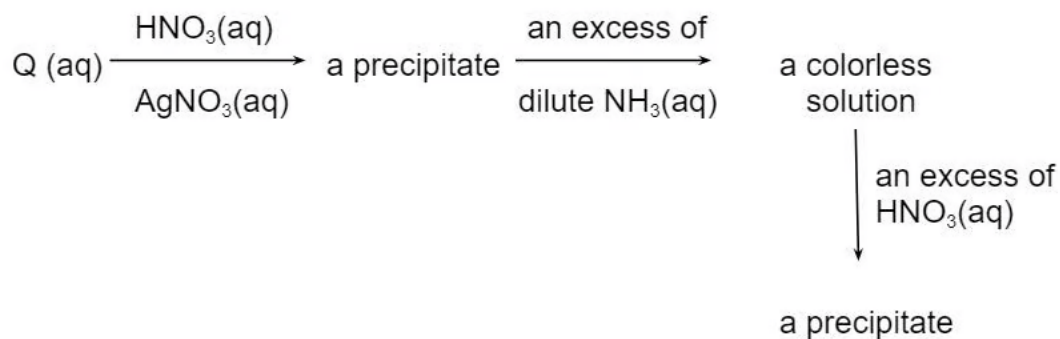
- A. Sodium sulfate
- B. Hydrogen sulfide
- C. Hydrogen astatide
- D. Astatine

[1 mark]

Question 6

Q is a salt of one of the Group 17 elements chlorine, bromine, iodine, or astatine.

The reaction scheme shows a series of reactions using a solution of **Q** as the starting reagent.



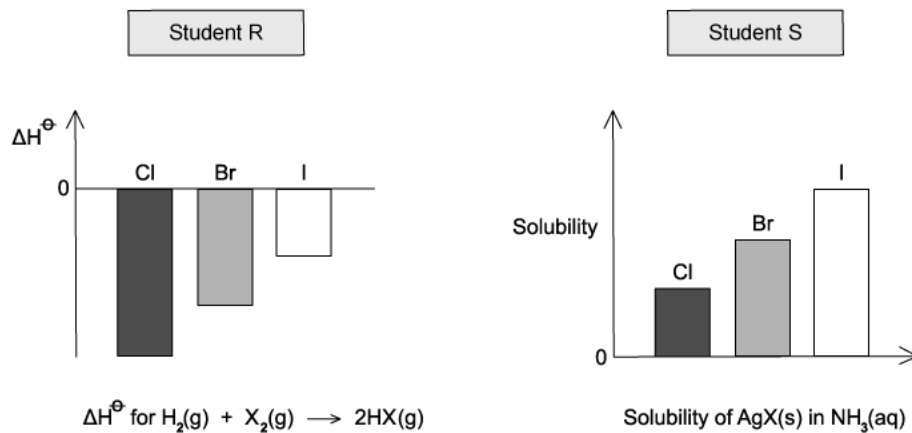
What could **Q** be?

- A. Sodium chloride
- B. Sodium bromide
- C. Potassium iodide
- D. Potassium astatide

[1 mark]

Question 7

Two students, R and S, were asked to draw bar charts to represent how some properties of the halogens and their compounds differ in magnitude. Their diagrams are shown.



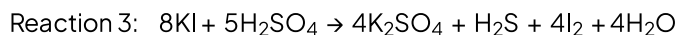
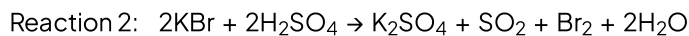
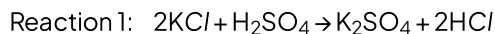
Which of the student's diagrams are correct?

- A. S only
- B. R only
- C. Both R and S
- D. Neither R nor S

[1 mark]

Question 8

Solid potassium halides react with concentrated sulfuric acid, according to the following equations.



What is the largest **change** in the oxidation number of sulfur in each of these reactions?

	Reaction 1	Reaction 2	Reaction 3
A	0	0	4
B	0	2	4
C	0	2	8
D	0	4	8

[1 mark]

Question 9

X, Y and Z represent different halogens. The table shows the results of nine experiments in which aqueous solutions of X_2 , Y_2 and Z_2 were separately added to separate aqueous solutions containing X^- , Y^- and Z^- ions.

	$\text{X}^- (\text{aq})$	$\text{Y}^- (\text{aq})$	$\text{Z}^- (\text{aq})$
$\text{X}_2 (\text{aq})$	No reaction	No reaction	No reaction
$\text{Y}_2 (\text{aq})$	X_2 formed	No reaction	Z_2 formed
$\text{Z}_2 (\text{aq})$	X_2 formed	No reaction	No reaction

Which row in the following table contains the ions X^- , Y^- and Z^- in order of their decreasing strength as reducing agents?

	Strongest → Weakest		
A	X^-	Y^-	Z^-
B	X^-	Z^-	Y^-
C	Y^-	Z^-	X^-
D	Z^-	X^-	Y^-

[1 mark]

Question 10

Which row represents the correct electrodes used to manufacture chlorine from brine in a diaphragm cell?

	Nature of cathode	Nature of anode
A	Titanium	Graphite
B	Titanium	Steel
C	Graphite	Titanium
D	Steel	Titanium

[1 mark]

Model Answers: Hard

1

The correct answer is **D** because:

- Concentrated sulfuric acid can act both as an acid and an oxidising agent.
- When it acts as an acid it produces hydrogen halide and if it was a halogen sodium salt, sodium hydrogen sulfate.
 - All the halogens react in this way.
- The reactions with sodium fluoride with both acids are this type of reaction.
- The reaction with sodium bromide and phosphoric acid is this type of reaction.
- When it acts as an oxidising agent, meaning the halide ion is acting as a reducing agent.
- Only bromine and iodine are strong enough reducing agents to reduce the concentrated sulfuric acid.
 - For example, in sodium bromide; the bromide ions are oxidised to bromine.
 - The bromide ions reduce the sulfuric acid to sulfur dioxide gas.
- The complete reaction takes place in this case:
 - $\text{H}_2\text{SO}_4 + 2\text{HBr} \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
- In this reaction with sodium bromide a yellow vapour of bromine is seen and steamy fumes of hydrogen bromide.
- Bromine is acting as a strong reducing agent and sulfuric acid a strong oxidising agent.

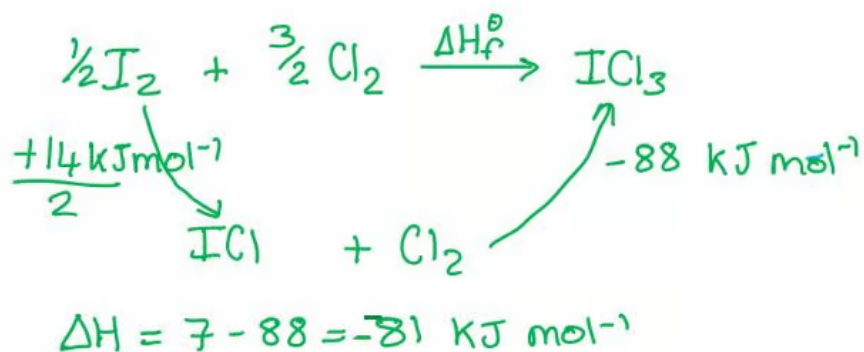
A & B are incorrect as phosphoric acid is not an oxidising agent and is acting as an acid.

C is incorrect as sulfuric acid is not a strong enough reducing agent to oxidise fluoride ions.

2

The correct answer is **B** because:

- The enthalpy change of formation ($\Delta H_f^\ominus$) is the enthalpy change when one mole of a compound is formed from its elements under standard conditions. The reactants and products must be in their standard states.
- To answer this question construct a Hess cycle and use the information given in the question to show the energy of the alternate routes.
- Remember that the definition is for one mole of the compound, so the equation needs to be balanced to produce one mole of ICl_3 .



3

The correct answer is **D** because:

- The test has to be done in solution, so the powders were added to water.
- When acidified silver nitrate is added a pale yellow colour is seen.
 - The pale yellow colour shows the presence of sodium iodide.
- When the concentrated ammonia was added the precipitate partially dissolved and leaves a darker yellow precipitate.
 - The darker yellow precipitate confirms the presence of sodium iodide.
 - The partial dissolving of the precipitate confirms the presences of sodium chloride.

A is incorrect as the precipitate would not partially dissolve when concentrated

aqueous ammonia was added.

B & C are incorrect as the precipitate would be a cream colour when aqueous silver nitrate was added.

4

The correct answer is **B** because:

- When potassium chloride is mixed with ammonia and silver nitrate, the reaction produces a colourless solution.
 - This is often carried out in two stages, stage one mixed with acidified silver nitrate to produce a white precipitate.
 - The ammonia is then added, and the white precipitate dissolves leaving a colourless solution.
 - $\text{AgNO}_3 + \text{KCl} \rightarrow \text{AgCl} + \text{KNO}_3$
- When potassium chloride is added to sulphuric acid steamy fumes of HCl are formed.
 - $\text{H}_2\text{SO}_4 + 2\text{KCl} \rightarrow \text{K}_2\text{SO}_4 + 2\text{HCl}$
- If the HCl is bubbled through potassium iodide, it will not react with the potassium iodide so the solution will be colourless.

5

The correct answer is **D** because:

- The reducing ability of the halide ions increases as you go down the group.

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5

The correct answer is **D** because:

- The reducing ability of the halide ions increases as you go down the group.
 - Astatine is at the bottom of Group 17.
- Concentrated sulfuric acid can act both as an acid and an oxidising agent.
- When it acts as an oxidising agent, it means the halide ion is acting as a reducing agent.
- The following reactions will take place in this case:
 - $\text{NaAt (s)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow \text{HAt (g)} + \text{NaHSO}_4 \text{ (s)}$
 - $2\text{NaAt (s)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow 2\text{HAt (g)} + \text{Na}_2\text{SO}_4 \text{ (s)}$
 - $2\text{HAt (s)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow \text{At}_2 \text{ (s)} + \text{SO}_2 \text{ (g)} + \text{H}_2\text{O (l)}$
 - $6\text{HAt (g)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow 3\text{At}_2 \text{ (s)} + \text{S (s)} + 3\text{H}_2\text{O (l)}$
 - $8\text{HAt (g)} + \text{H}_2\text{SO}_4 \text{ (aq)} \rightarrow 4\text{At}_2 \text{ (s)} + \text{H}_2\text{S (g)} + 4\text{H}_2\text{O (l)}$
- In the third reaction shown At_2 is produced the astatine has been oxidised in the following reaction.



- In this reaction HAt is a reducing agent, H_2SO_4 is an oxidising agent.

6

The correct answer is **A** because:

- The results of the tests for halide ions and acidified silver nitrate and then ammonia:

Halide ion	Precipitate $\text{HNO}_3 + \text{AgNO}_3$	Dilute NH_3	Excess HNO_3
Cl^-	White	Dissolves	White precipitate
Br^-	Cream	Stays insoluble	Stays insoluble
I^-	Pale yellow	Stays insoluble	Stays insoluble
At^-	Yellow	Stays insoluble	Stays insoluble

- Compound Q must be **A** sodium chloride as this is the only halide ion that will:
 - o Produce a precipitate with acidified silver nitrate.
 - o Colourless solution with ammonia.
 - o Form a precipitate with nitric acid.

7

The correct answer is **B** because:

- The bar graph drawn by student R shows the enthalpy change when a reaction takes place between a halide ion and hydrogen to make hydrogen halides.
- The enthalpy change of this reaction gets less exothermic as you go down Group 17.
- This is due to the reactivity of the Group 17 elements; they are more reactive at the top of the group.
- The reactions of the halogens with hydrogen gas get less explosive as you go down the group.

- Less energy is released from the reaction as in the graph by student R.

The graph drawn by student S is incorrect as the solubility of the silver halides in ammonia decreases as you go down the group.

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Cl^-	White	Dissolves
Br^-	Cream	Stays insoluble
I^-	Pale yellow	Stays insoluble

8

The correct answer is **C** because:

- In **reaction 1** there is no change in the oxidation state of sulfur.
 - Both hydrogen and potassium have an oxidation state of +1 meaning that the sulfur oxidation state in these compounds is +6.
- In **reaction 2** the bromide ions reduce the sulfuric acid to sulfur dioxide gas, this decreases the oxidation state of the sulfur from +6 in the sulfuric acid to +4 in the sulfur dioxide.
 - A change in the oxidation state of 2.
- In **reaction 3** the reduction of sulfuric acid (oxidation state +6) is more complex.
 - The first stage is to sulfur dioxide (sulfur oxidation state +4).
 - Then to sulfur (oxidation state 0)

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- The first stage is to sulfur dioxide (sulfur oxidation state +4).
- Then to sulfur (oxidation state 0).
- And then to hydrogen sulfide (sulfur oxidation state -2).
- This makes the greatest change in oxidation state in this reaction 8 (from +6 to -2).

9

The correct answer is **B** because:

- A more reactive halogen can displace a less reactive halogen from a halide solution of the less reactive halogen.
- If the solutions are mixed cyclohexane after the reactions, then the colours of the halogens can be more clearly seen.

Halogen	Reaction with Cl^-	Reaction with Br^-	Reaction with I^-
Cl_2	None	Brown colour	Purple colour
Br_2	None	None	Purple colour
I_2	None	None	None

- The reducing power of the halogens increases going down Group 17.
- The results in the experiment show that:
 - $\text{X}_2 = \text{I}_2$
 - $\text{Y}_2 = \text{Cl}_2$
 - $\text{Z}_2 = \text{Br}_2$

	$\text{I}^- (\text{aq})$	$\text{Cl}^- (\text{aq})$	$\text{Br}^- (\text{aq})$
$\text{I}_2 (\text{aq})$	No reaction	No reaction	No reaction
$\text{Cl}_2 (\text{aq})$	I_2 formed	No reaction	Br_2 formed
$\text{Br}_2 (\text{aq})$	I_2 formed	No reaction	No reaction

- Therefore, the order of decreasing reducing power is X, Z, Y – option **B**.

10

The correct answer is **D** because:

- Sodium chloride solution contains:
 - Na^+ ions
 - Cl^- ions
 - H^+ ions
 - OH^- ions
- Hydrogen gas is released from a steel cathode.
- Chlorine gas is released from a titanium anode.

